Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy across primary energy, electricity generation, hydropower, wind, solar, and regional trans

~14%
global primary energy from renewables

~30%
global electricity from renewables

~9,600 TWh
renewable generation across listed technologies

Audience: policy analysts and energy sector strategists

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

75% of global GHG emissions still come from fossil fuels

Fossil fuel dependence remains both a climate and public-health problem.

Climate pressure

75%

of global greenhouse gas emissions come from burni

Health burden

5M+

premature deaths per year are linked to local air pollu

System lock-in

Since Industrial Revolution

most countries’ energy systems have become dominated

lication: rapidly shift energy systems toward low-carbon sources — renewable technologies and nuclear — while reducing air pollutio

Renewables now supply ~14% of global primary energy

Primary energy includes electricity, transport, and heating; data use the substitution method.

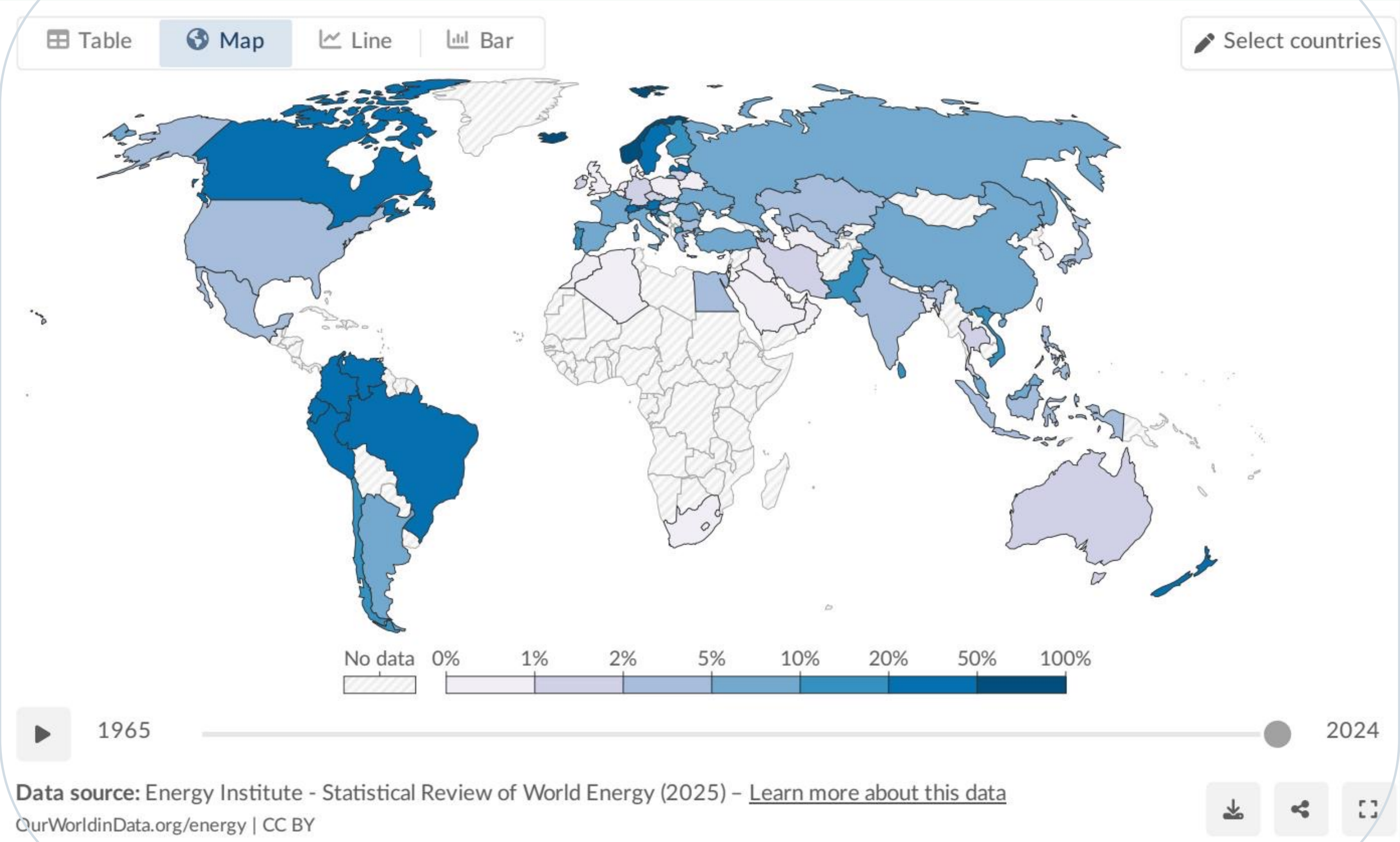


Figure: Share of primary energy from renewables by country, 2024

~14%

of global primary energy now comes from renewable technologies

1/7

approximately one-seventh of primary energy is renewable

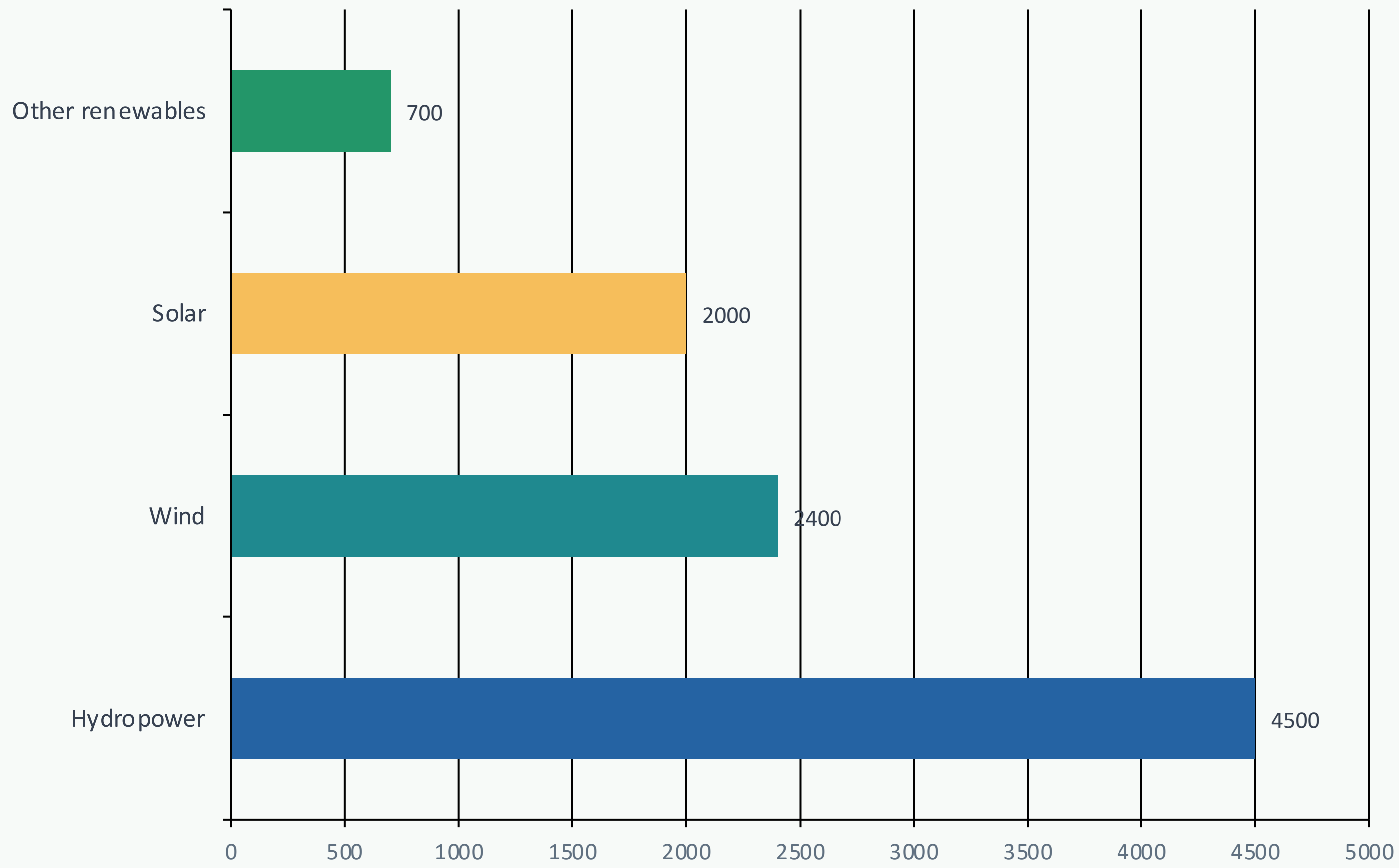
Why the gap matters

Transport and heating remain harder to decarbonize, so renewables' share of total primary energy is much lower than their electricity share.

High-renewable primary energy shares appear in countries such as Norway, Brazil, and New Zealand; fossil-fuel producers in parts of the Middle East and Asia remain below 5%.

Wind and solar now nearly match hydropower’s renewable output

Renewable electricity generation by technology, 2024; generation values from the brief.



Approx. global generation (TWh)

Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other renewables	~700	~7%

Acceleration signal

Wind and solar have accelerated sharply since around 2010; together they produce ~4,400 TWh, nearly hydropower’s ~4,500 TWh.

Hydropower remains the single largest renewable source, but wind and solar are driving the growth story.

Our World in Data — Renewable Energy; Ember (2026); Energy Institute (2025)

Almost one-third of global electricity now comes from renewables

Electricity is where renewables have made the most visible progress.

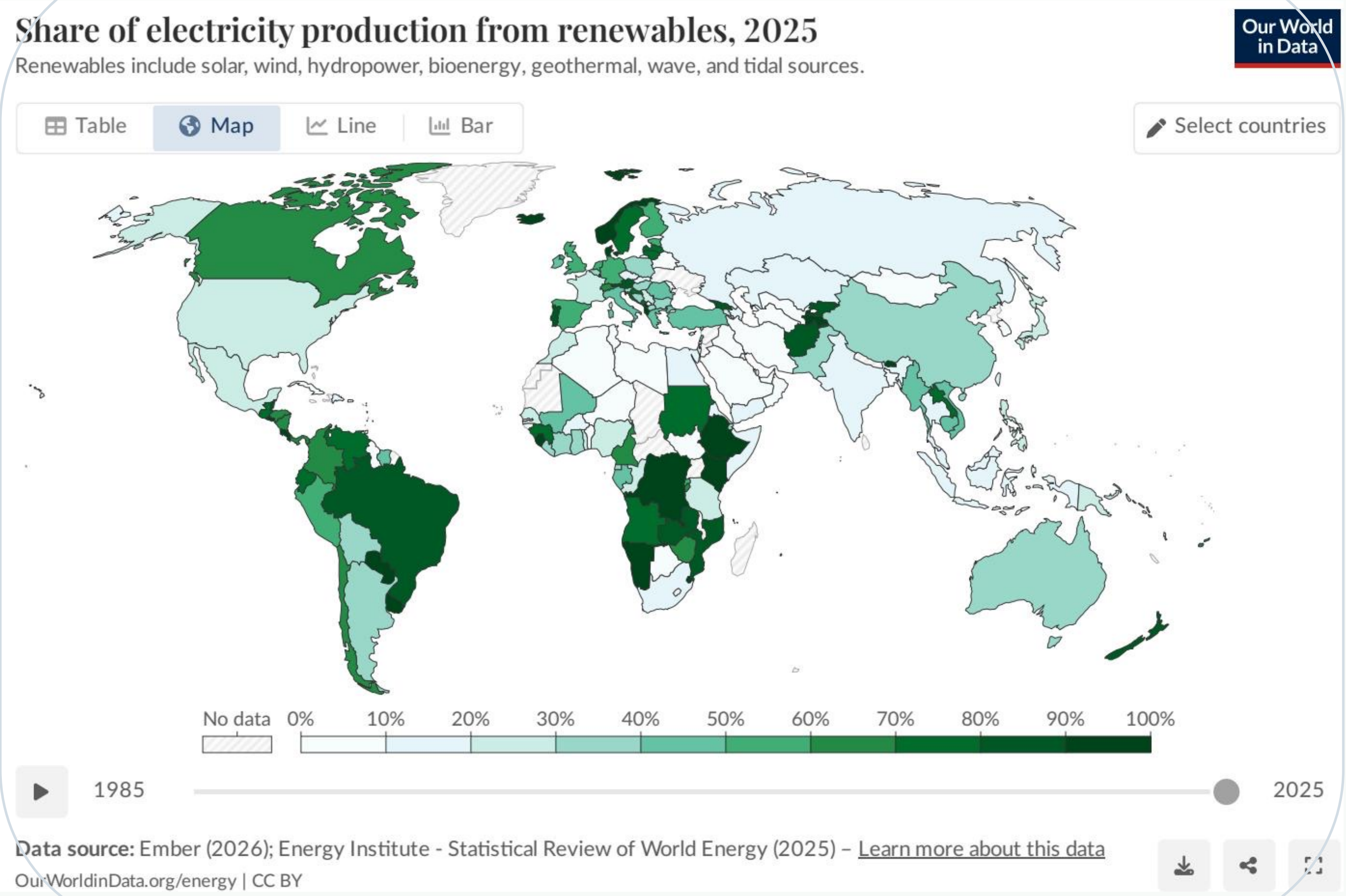


Figure: Share of electricity from renewables by country, 2025

Country / region	Renewable share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

~30%
world average share of electricity from renewables

60–90%+
leaders include Brazil, Canada, and Nordic countries

Several African and Middle Eastern countries remain below 10%, reflecting fossil fuel dependence and limited renewable investment.

Hydropower remains the largest renewable source at ~50% of generation

A century-old technology still anchors renewable electricity systems in several large markets.

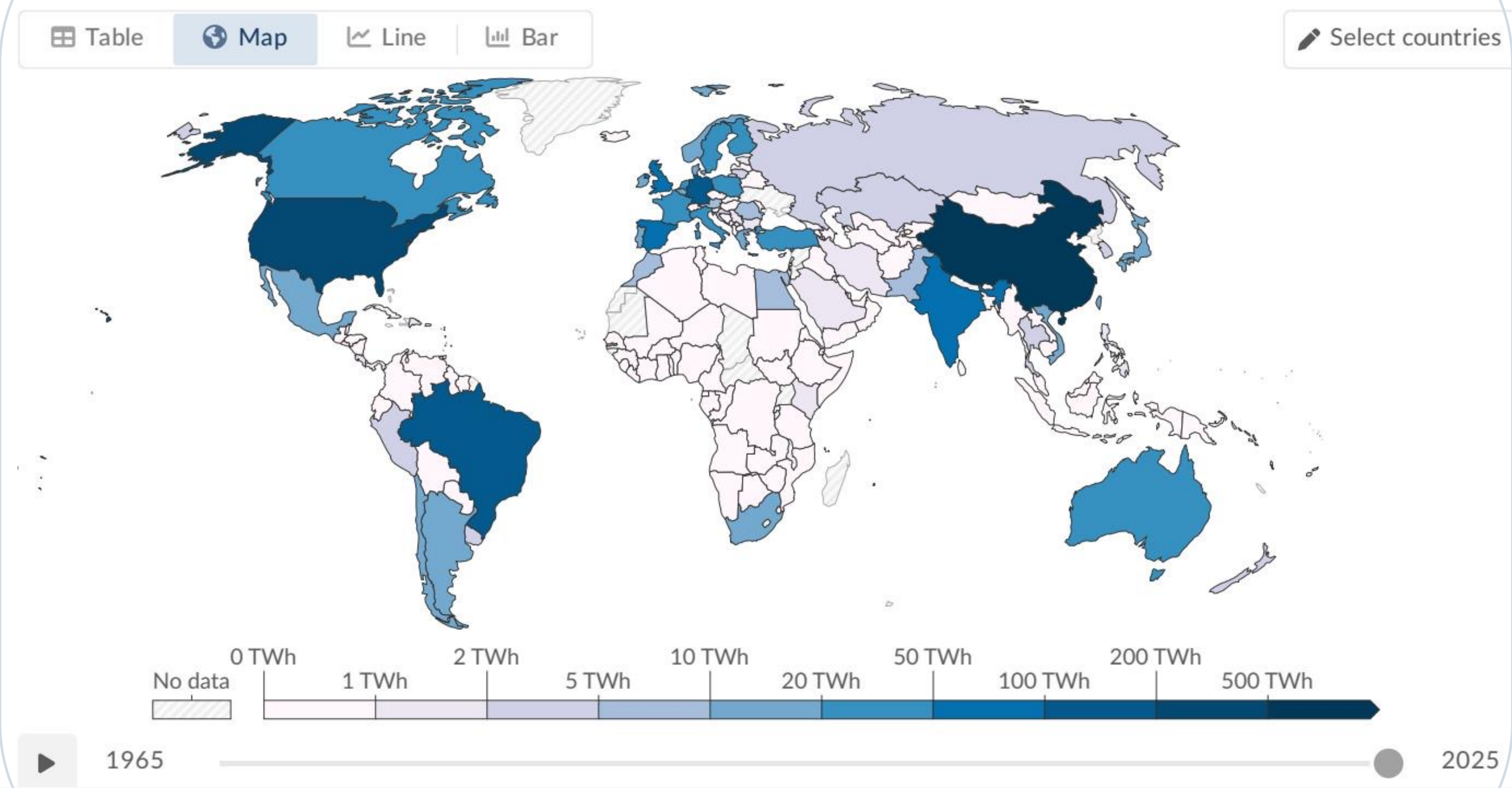


Figure: Hydropower generation by country, 2025

~4,500 TWh

approx. global hydropower generation in 2024

~47%

share of renewable electricity generation

- Top producers include China, Brazil, and Canada.
- Hydropower benefits countries with favorable river systems a
- Growth potential is more limited than wind and solar, even as

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Wind and solar are the fastest-growing energy technologies in history

Generation data — not installed capacity — show the scale of their contribution.

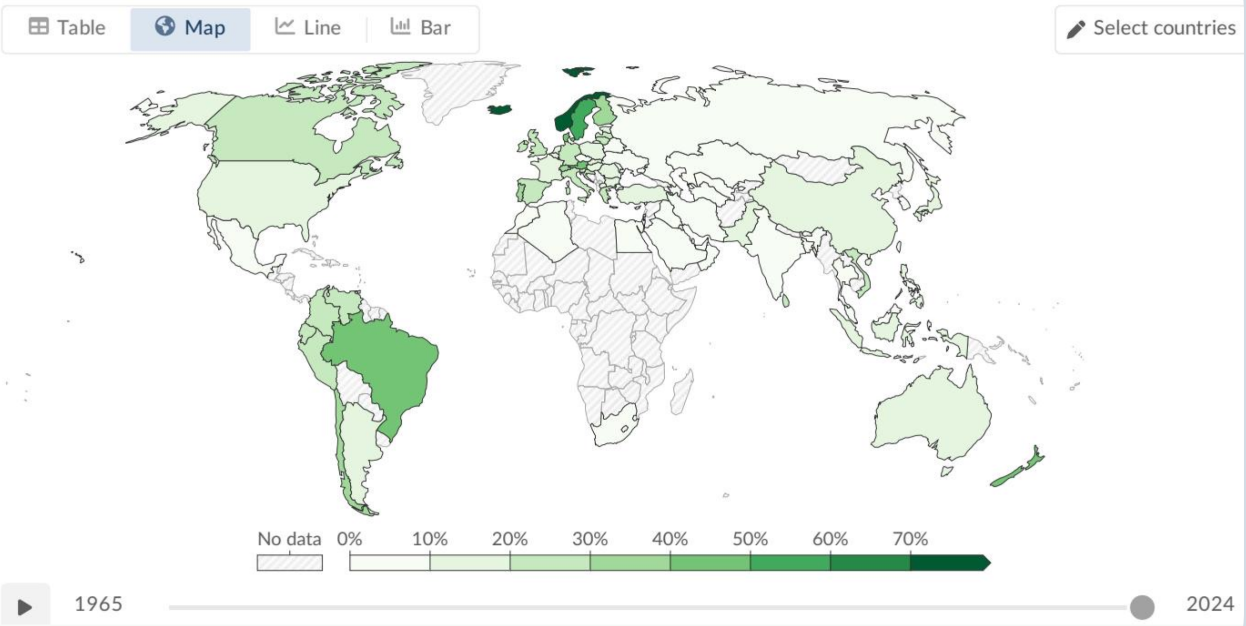
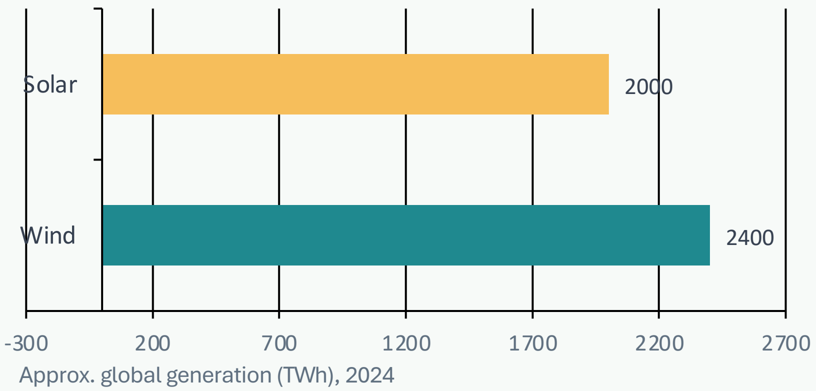


Figure: Wind power generation by country, 2025

Wind vs. solar generation and renewable share



Technology	Generation	Share
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%

Solar is the fastest-growing source, with dramatic growth since 2010 driven by falling costs and policy support.

Together: ~4,400 TWh — nearly matching hydropower's ~4,500 TWh output.

Stark regional gaps: leaders exceed 60–90%, while laggards remain below 10%

The transition is uneven across resources, investment, and infrastructure.

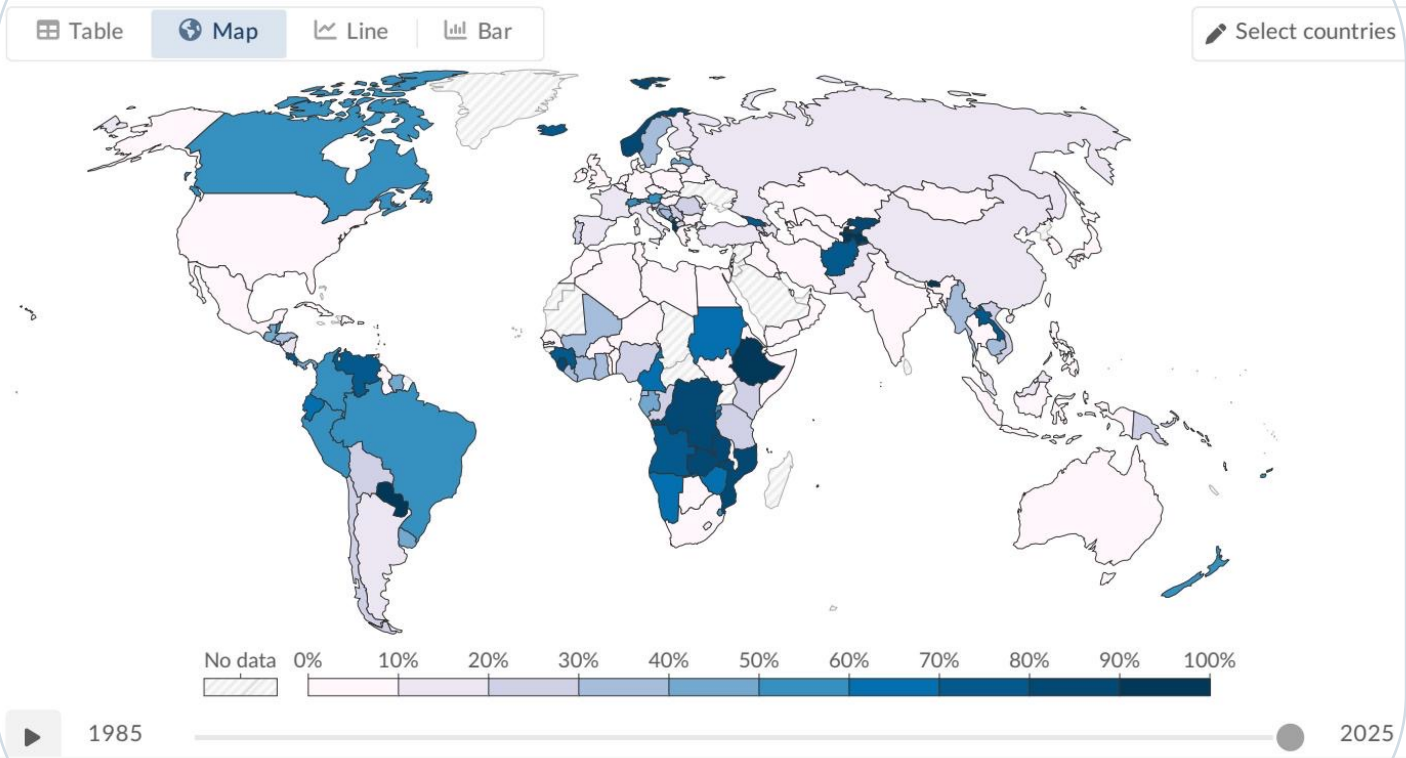


Figure: Share of electricity from hydropower by country, 2025

Renewable leaders

Scandinavia, Brazil, and Canada benefit from abundant hydropower; Brazil (~85%), Canada (~65%), and Norway (~98%) are high-renewable electricity systems.

Investment pathway

Denmark and Germany have invested heavily in wind and solar; Germany is around ~55% renewable electricity.

Access and dependence gap

Several African and Middle Eastern countries are below 10%; Saudi Arabia remains below 2% in renewable electricity.

Sub-Saharan Africa has minimal renewable electricity infrastructure, while many Middle Eastern nations remain almost entirely fossil-fuel dependent.

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Key takeaways: renewables are growing fast, but the transition must accelerate

~14% / ~30%

Renewables supply ~14% of global primary energy and almost one

Hydro still leads

Hydropower generates roughly half of all renewable electricity, le

Wind + solar catching up

Wind (~2,400 TWh) and solar (~2,000 TWh) together now nearly

Regional gaps persist

Scandinavia and Latin America lead, while much of the Middle Ea

Beyond electricity

Transport and heating lag behind power-sector decarbonization, keeping primary-energy progress

Forward look: accelerate deployment where renewables are already scaling, and close infrastructure and investment gaps where fossil dependence remains high.

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)